

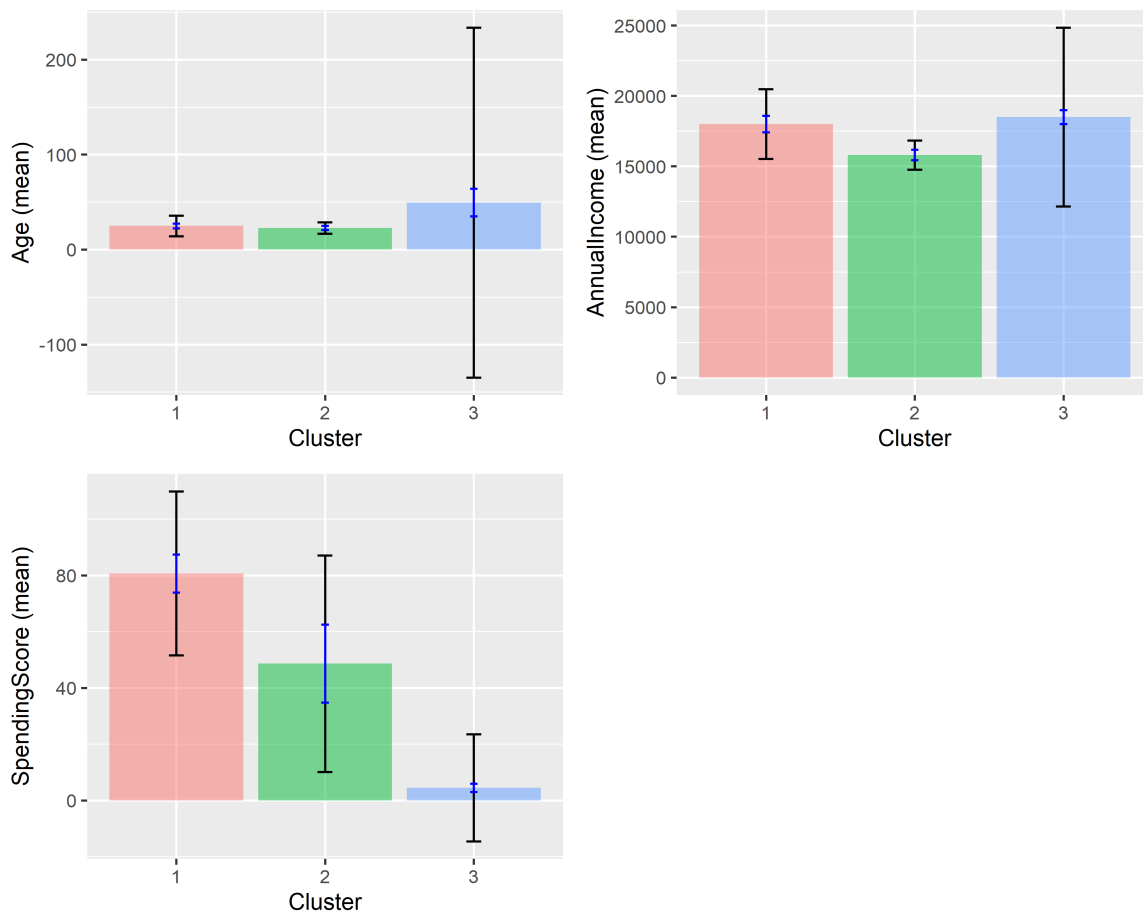
Interactive Business Analytics Using R and Shiny

The Radiant Package

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Interactive Business Analytics Dashboard



Presentation Outline

1. What is Interactive Business Analytics?
 2. Traditional vs Interactive Analytics
 3. Introduction to Radiant
 4. The Shiny Framework
 5. Live Demo: Customer Segmentation
 6. Building Custom Apps & Summary
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What is Interactive Business Analytics?

Definition: Dynamic tools that allow business users to explore data, change parameters, and see results in real time.

Real-Time Responsiveness

- Adjust a slider - plot updates instantly
- No waiting, no refresh button
- Immediate visual feedback

User-Driven Exploration

- Users control what they see
- Ask their own questions
- Discover patterns themselves

Accessibility

- Point-and-click interface
- No coding required
- Reduces dependency on analysts

Reproducibility

- All actions generate underlying R code
- Results can be verified and repeated
- Decisions are documented

Traditional vs Interactive Analytics

Feature	Traditional (Static)	Interactive (Dynamic)
Who explores?	Analyst creates report	User explores directly
Parameters	Fixed	Adjustable in real time
Turnaround	Days / hours	Instant
Insights	One-time	Continuous

Why does this matter?

- **Speed:** No waiting for analyst reports
- **Exploration:** Users answer their own questions

- **Scenarios:** Test multiple “what-if” situations
 - **Democratisation:** Analytics for everyone
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What is Radiant?

Radiant is a platform-independent, browser-based interface for business analytics in R.

Key Features

- **Data:** Import, view, transform
- **Basics:** Explore, visualise
- **Model:** Regression, prediction
- **Multivariate:** Clustering, PCA
- **Report:** Export reproducible output

Installing and Launching

```
install.packages("radiant")  
library(radiant)  
radiant()
```

Why Radiant?

- Built on R analytics + Shiny interactivity
 - Point-and-click, no coding required
 - Generates reproducible R code automatically
 - Free and open-source
 - Runs in any browser
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The Shiny Framework

Shiny is an R framework for building interactive web applications.

Core concept: Reactive Programming - outputs automatically update when inputs change.

```
library(shiny)

ui <- fluidPage(
  sliderInput("threshold", "Sales Threshold:",
             min = 0, max = 100000, value = 50000),
  plotOutput("salesPlot")
)

server <- function(input, output) {
  output$salesPlot <- renderPlot({
    # Reacts automatically when input$threshold changes
  })
}

shinyApp(ui = ui, server = server)
```

- **ui** - controls what the user **sees**
- **server** - controls how the app **responds**
- **shinyApp()** - combines both parts

Demo: Customer Segmentation Setup

Business question: How can we segment customers by age, income, and spending behaviour?

Step 1 - Launch Radiant

```
library(radiant)
radiant()
```

Step 2 - Load data: Data → Manage → Load data → select `customers_data.csv` → click Load

Step 3 - Verify dataset: Data → View → confirm columns are present:

Age	AnnualIncome	SpendingScore
19	15000	39
21	15000	81

Demo: Running the Analysis

Step 4-5 - Configure K-Means Clustering

Multivariate → Cluster → select variables: **Age, AnnualIncome, SpendingScore** → Method: **K-means** → k = **3** → click **Estimate**

Step 6 - Observe the interactive output

After running, Radiant displays:

- Cluster results table
- Cluster sizes and means
- Visual plots
- Summary statistics

This is Shiny's reactivity in action; change k from 3 to 4 and all outputs update instantly.

Interpreting the Customer Segments

Segment 1 High-Value Customers

- High income
- High spending score

Strong purchasing power, already highly engaged. Priority for loyalty programmes.

Segment 2 Budget-Conscious Customers

- Lower income
- Lower/moderate spending

Price-sensitive group. Respond well to discounts and value-for-money offers.

Segment 3 Growth Potential Customers

- Good income
- Lower spending score

Ability to spend more but not yet engaged. Target with personalised campaigns.

Building Custom Shiny Apps

Why go beyond Radiant?

- Specific business workflows
- Company branding and databases
- Complex multi-step logic
- User-friendly interfaces for non-R users

Deployment Options

Platform	Best For	Cost
shinyapps.io	Quick prototypes and sharing	Free tier available
Shiny Server	Internal company use	Free (open source)
RStudio Connect	Enterprise deployment	Commercial

Radiant gives you the interface. Shiny gives you the freedom to build your own.

Class Exercise

Question:

Which R package would you use first to create an interactive business analytics dashboard: Radiant or Shiny and why?

Think about:

- Ease of use vs customisation
- Business purpose vs technical flexibility
- Who is the end user?
- What does your organisation already know?

There is no single correct answer. The goal is to justify your choice.

Summary

Key Takeaways

- Interactive analytics empowers users to explore data without coding
- **Radiant** = R + Shiny, packaged as a business analytics interface
- **Shiny** = the reactive framework that powers Radiant and custom apps
- Customer segmentation is a practical, high-value application
- Both tools are free, open-source, and run in the browser

The Core Relationship

```
# R
#   Shiny (reactive web framework)
#   Radiant (business analytics UI)
#   Your Custom Apps
```

Start with Radiant. Build with Shiny.

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Thank You